OXFORD

INTERNATIONAL AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

INTERNATIONAL AS

CHEMISTRY (9620)

Unit 2: Organic 1 and Physical 1

Monday 22 May 2017 Morning Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

Do not write outside the box

Answer **all** questions in the spaces provided.

0 1 This question is about pentadecane, $C_{15}H_{32}$

0 1 . 1 Write an equation to show the incomplete combustion of pentadecane to form carbon monoxide as the only carbon-containing product.

[1 mark]

0 1 . 2 Nitrogen monoxide is also formed during the combustion of pentadecane in motor vehicles.

Describe how nitrogen monoxide is formed.

[2 marks]

0 1 . 3 Write an equation to show how nitrogen monoxide is removed by a catalytic converter.

[1 mark]

Do not write outside the box

0 1 . 4 When pentadecane is used as a fuel in industry, sulfur-containing impurities cause the formation of sulfur dioxide and other combustion products.

Explain how sulfur dioxide can be removed from these other combustion products. Include an equation in your answer.

[2 marks]

0 1 . 5 Pentadecane can be converted into other useful molecules by cracking.

State the conditions needed for the cracking of pentadecane to form a high proportion of aromatic hydrocarbons.

[2 marks]

0 1 . 6 Write an equation showing one molecule of pentadecane being cracked into hexane, propene and one other product.

[1 mark]

9

Turn over for the next question

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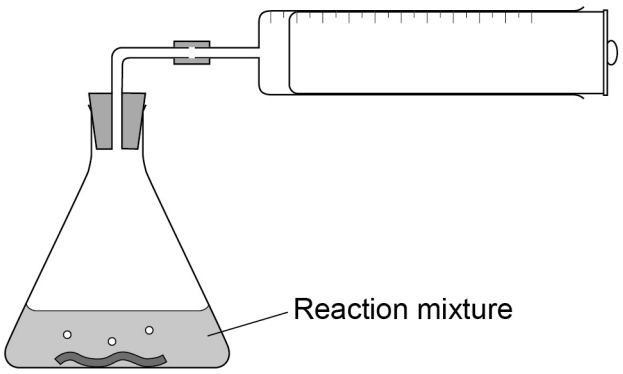
0 2 A student investigated the reaction between magnesium and dilute hydrochloric acid.

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

The student placed 105 mg of magnesium into a conical flask and added an excess of dilute hydrochloric acid.

The student then connected the conical flask to a gas syringe as shown in **Figure 1** and recorded the volume of gas collected every 20 seconds.

Figure 1



0 2 . 1 Calculate the minimum volume, in cm^3, of 0.500 mol dm^{-3} hydrochloric acid needed to react with 105 mg of magnesium.

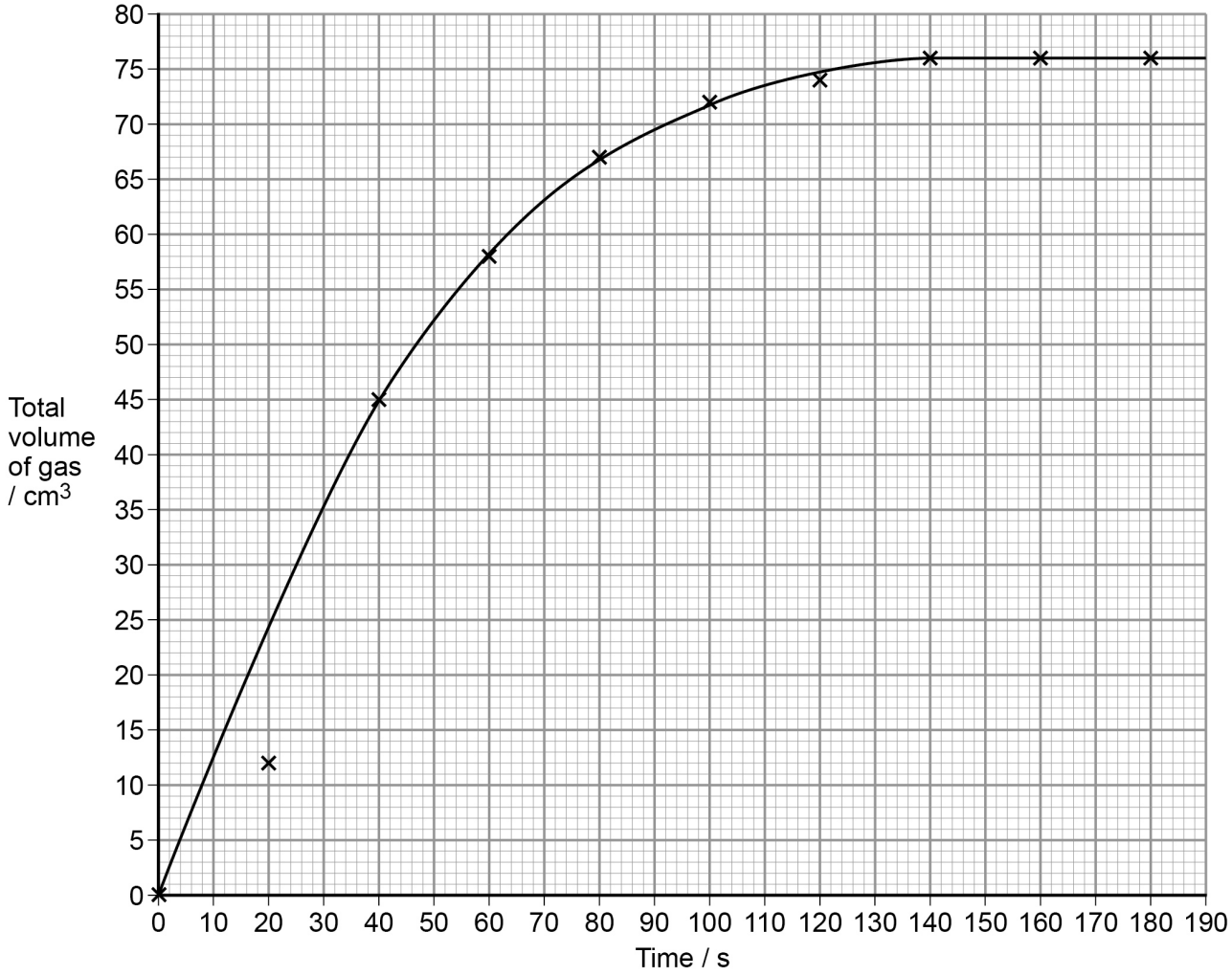
[3 marks]

Volume = _____ cm^3

Do not write outside the box

0 2 . 2 A graph of total volume of gas against time was plotted. This is shown in **Figure 2**.

Figure 2



Draw a tangent to the curve at 50 seconds.

Use the tangent to determine the rate of reaction, in $cm^3\ s^{-1}$, at 50 seconds.
Show your working.

[2 marks]

Rate of reaction = _____ $cm^3\ s^{-1}$

Question 2 continues on the next page

Do not write outside the box

0 2 . 3 Suggest a practical reason why the total volume of gas collected is lower than expected.

[1 mark]

0 2 . 4 Explain the effect of increasing the concentration of dilute hydrochloric acid on the time taken to collect 50 cm^3 of hydrogen gas.

[2 marks]

8

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0 3 Propan-2-ol undergoes combustion when it is ignited.

0 3 . 1 Define the term standard enthalpy of combustion.

[2 marks]

0 3 . 2 An equation for the complete combustion of gaseous propan-2-ol is shown.

$CH_3CH(OH)CH_3(g) + 4\frac{1}{2}O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(g) \quad \Delta H = -1886 \text{ kJ mol}^{-1}$

Use the enthalpy change for the reaction and the mean bond enthalpy data shown in **Table 1** to calculate the value of the bond enthalpy for the C—O bond in propan-2-ol.

Table 1

	C—H	C—C	O—H	O=O	C=O
Mean bond enthalpy / kJ mol^{-1}	412	348	463	496	805

[3 marks]

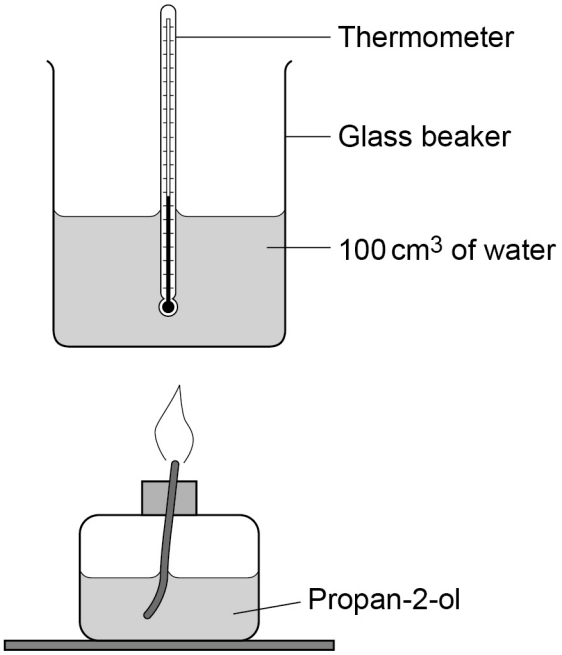
Bond enthalpy = _____ kJ mol^{-1}

Question 3 continues on the next page

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A pure sample of propan-2-ol was placed in a spirit burner and positioned under a glass beaker containing 100 cm^3 of water. This is shown in **Figure 3**. The wick of the spirit burner was ignited and the fuel was allowed to burn for several minutes before the flame was extinguished.

Figure 3



The results for the experiment are shown in **Table 2**.

Table 2

Initial temperature of water / °C	19.0
Final temperature of water / °C	37.0
Initial mass of spirit burner and propan-2-ol / g	187.797
Final mass of spirit burner and propan-2-ol / g	187.231

Do not write outside the box

0 3 . 3 Calculate a value, in kJ mol^{-1}, for the enthalpy of combustion of this sample of propan-2-ol.

The specific heat capacity of water = 4.18 J K^{-1} g^{-1}.

[4 marks]

Enthalpy of combustion = _____ kJ mol^{-1}

0 3 . 4 Suggest how the apparatus that the student used could be adjusted to improve the accuracy of the experiment.

[1 mark]

_____

_____

_____

0 3 . 5 A student repeated this experiment and obtained a temperature rise of 16.0 °C. The uncertainty in each of the temperature readings from the thermometer in this experiment was ±0.5 °C.

Calculate the percentage uncertainty for the temperature change in this experiment.

[1 mark]

Percentage uncertainty = _____ %

11

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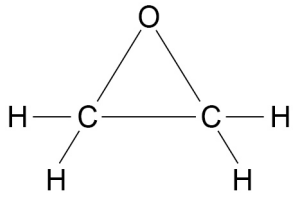
0 4 Epoxyethane is formed by the reaction of ethene with oxygen.

0 4 . 1 Identify the catalyst for this reaction.

[1 mark]

0 4 . 2 The displayed formula for epoxyethane is shown in **Figure 4**.

Figure 4



Explain, by reference to its structure, why epoxyethane is a reactive molecule.

[2 marks]

Do not write outside the box

0 4 . 3 In acidic conditions, epoxyethane reacts with water to form ethane-1,2-diol.

Outline a mechanism for this reaction.

[4 marks]

0 4 . 4 One molecule of epoxyethane reacts with one molecule of an alcohol to form a compound with M_r of 90.0. Analysis showed that this compound contained 53.3% carbon, 11.1% hydrogen and 35.6% oxygen by mass.

Calculate the molecular formula of the compound formed.
Show your working.

[3 marks]

Molecular formula of compound = _____

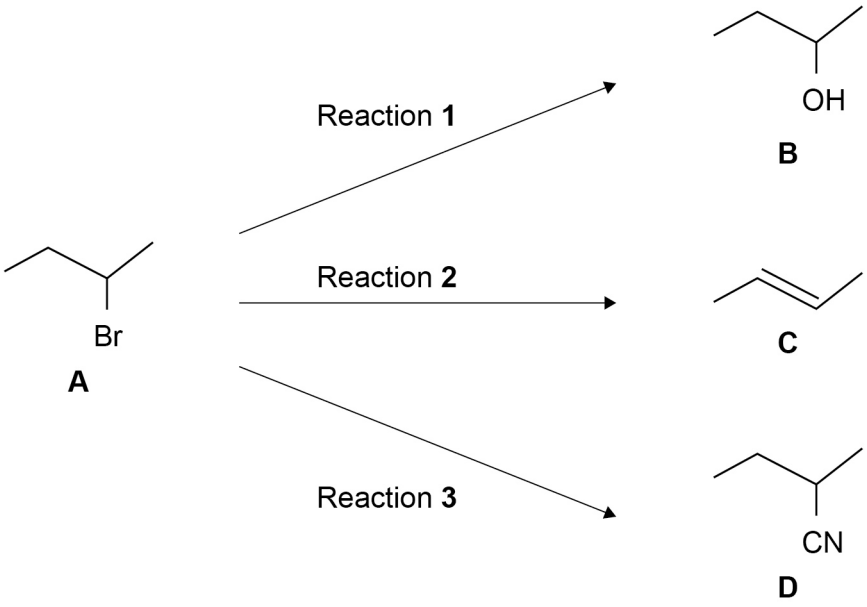
10

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0 5 Halogenoalkanes are important materials for making many different organic compounds.

Consider the following reactions of compound **A** to form compounds **B**, **C** and **D**, which are shown in **Figure 5**.

Figure 5



0 5 . 1 Give the IUPAC name for compound **D**.

[1 mark]

Do not write outside the box

0 5 . 2 Aqueous sodium hydroxide is the reagent used in reaction **1**.

Name and outline the mechanism for this reaction.

[3 marks]

Name _____

Mechanism

0 5 . 3 Ethanolic sodium hydroxide is used in reaction **2**.

State the role of sodium hydroxide in reaction **2** and name this type of reaction.

[2 marks]

Role _____

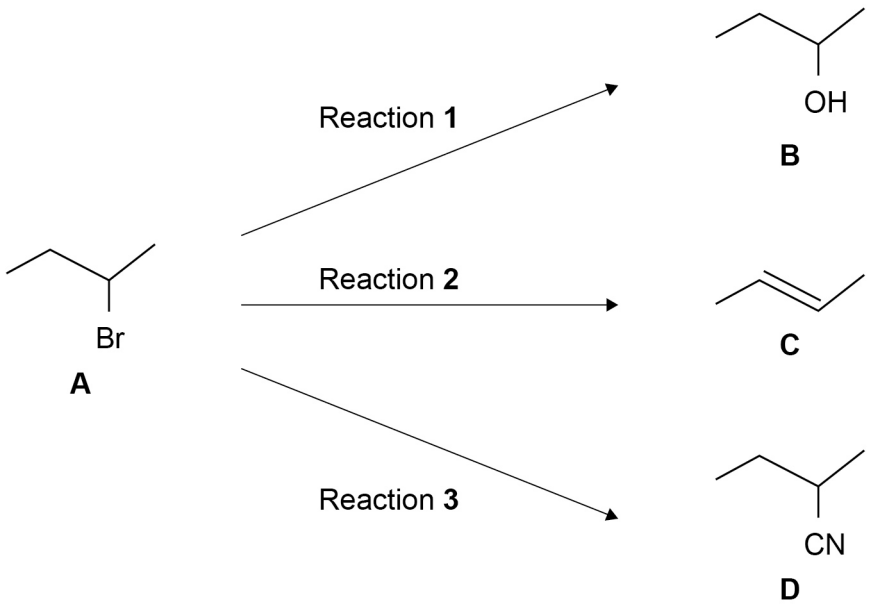
Type of reaction _____

Question 5 continues on the next page

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Figure 5 is repeated to help you answer the following questions.

Figure 5



0 5 . 4 Explain how the rate of reaction between 2-chlorobutane and aqueous sodium hydroxide compares with the rate of reaction between 2-iodobutane and aqueous sodium hydroxide. You should assume that both reactions are carried out under the same conditions.

[2 marks]

Rate of 2-chlorobutane compared with 2-iodobutane

Explanation

0 5 . 5 Infrared spectroscopy can be used to distinguish between compounds **A**, **B**, **C** and **D**.

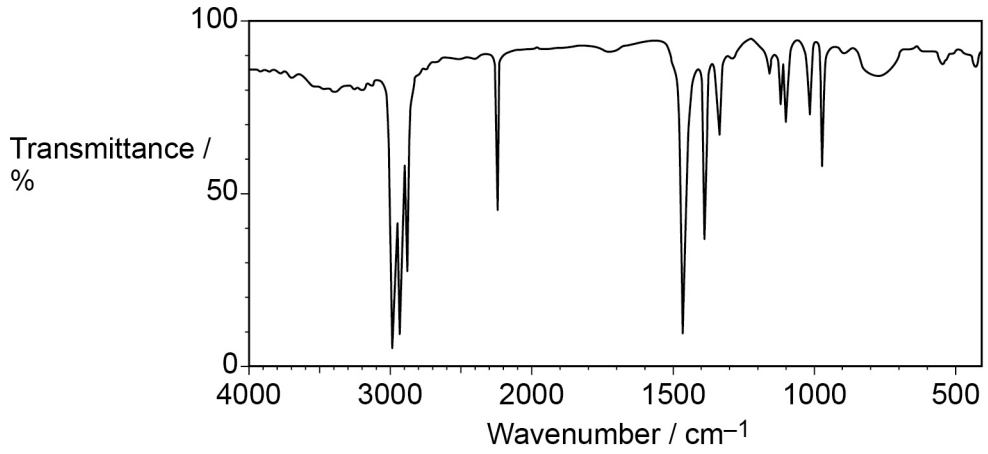
The infrared spectra of three of these compounds are shown in **Figure 6** on page 15.

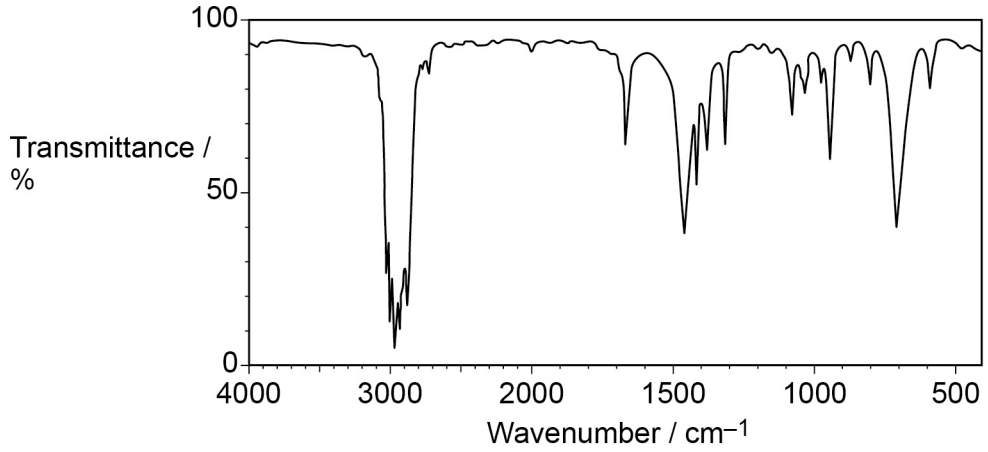
Identify which compound, **A**, **B**, **C** or **D**, produces each spectrum.
Write the letter of each compound in the box next to its spectrum in **Figure 6**.

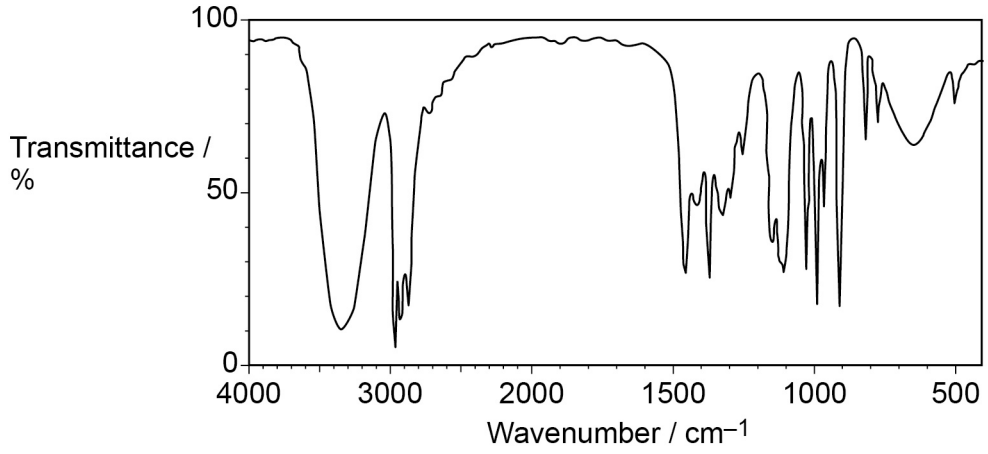
[3 marks]

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Figure 6







11

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0 6 This question is about alkenes.

0 6 . 1 Outline the mechanism for the reaction that occurs when butan-1-ol is heated with concentrated sulfuric acid to form but-1-ene.

[4 marks]

0 6 . 2 State the role of sulfuric acid in the reaction in Question **06.1**

[1 mark]

0 6 . 3 But-1-ene is a monomer in addition polymerisation reactions.

Draw the structure of the repeating unit formed during the polymerisation of but-1-ene.

[1 mark]

Do not write outside the box

0 6 . 4 Explain why the products of addition polymerisation are less reactive than the alkene monomers.

[1 mark]

_____

_____

_____

7

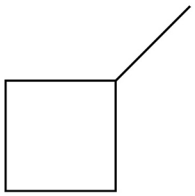
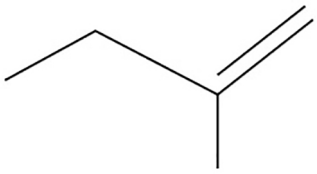
Turn over for the next question

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0 7 The molecular formula C_5H_{10} represents several different isomers.

0 7 . 1 **Table 3** contains information about three isomers of C_5H_{10}

Table 3

Isomer	Name	Skeletal formula
X	Pent-1-ene	
Y		
Z		

Complete **Table 3** for the three isomers of C_5H_{10}

[3 marks]

0 7 . 2 Describe a chemical test to distinguish between isomer **Y** and isomer **Z**.
State what would be observed with each isomer.

[3 marks]

Test _____

Observation with **Y** _____

Observation with **Z** _____

Do not write outside the box

0 7 . 3 Pent-2-ene shows *E*–*Z* isomerism.

Draw the skeletal formula of *E*-pent-2-ene.
Justify your answer using Cahn-Ingold-Prelog (CIP) priority rules.

[3 marks]

0 7 . 4 State why pent-1-ene does **not** show *E*–*Z* isomerism.

[1 mark]

10

Turn over for the next question

Do not write outside the box

0 8 In the reaction of propene with HBr, two possible intermediates are formed.

0 8 . 1 Name the mechanism for this reaction.

[1 mark]

0 8 . 2 Identify the major product formed in the reaction.
Explain why this is the major product.

[3 marks]

Major product

Explanation

4

END OF QUESTIONS

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